Simon J. Bloch

simon.j.bloch@gmail.com • Brooklyn, NY • 646-265-2202 • github.com/s-j-b

Software Engineer | Distributed Systems • AI Tools • Data Infrastructure

9+ years turning complex, ambiguous problems into working production systems.
Expert Python, applied AI, and mathematics. Deep grasp of underlying abstractions.

EXPERIENCE

Enlightapp — Lead Software Engineer *Mar 2024 – Present*

- Architected and shipped AI-powered "Classroom Community" feature: real-time multiplayer engagement system with LLM-driven feedback loops (Firebase, Next.js, Supabase, OpenAI API)
- Led and mentored international engineering team; established CI/CD pipeline, code review protocol, and on-call practices from scratch
- Redesigned user auth and onboarding stack for security and accessibility (OAuth 2.0, Edlink, Next.js)

Freelance / City Bureau — Software Engineer *Jan 2022 – Feb 2024*

- Built geospatial data visualization platform for research org: time-dependent shapefile infrastructure with hierarchical spatial aggregations over ~10M row PostGIS dataset (Python, Mapbox, D3.js, PostGIS)
- Developed and maintained Python web scrapers and scheduling systems for civic data pipeline (City Bureau)

Gro Intelligence — Software Engineer *Jun 2017 – Sep 2021*

- Designed and productionized geospatial crop mask raster pipeline (Python, GEE, AWS) [see **Selected Projects**]
- Led architecture redesign of search and indexing system over 100K+ entity knowledge graph; reduced query and indexing latency by order of magnitude (Python, Apache Solr)
- Spearheaded zero-downtime migration of user data to new schema (PostgreSQL); developed and owned DevOps practices, on-call rotations, and incident response; ran architecture training and hiring interviews

Carnegie Mellon Robotics Institute — Research Engineer Intern *Jun – Aug 2016*

- Built probabilistic perceptual model and 3D spatial mapping infrastructure for a state-of-the-art sensor designed for deployment to Jupiter's icy moons (C++, Octomap)

Google — Software Engineering Intern *May – Aug 2015*

- Chrome Media: designed and implemented noise-filtering beamforming layer of audio-processing pipeline (C++)

SELECTED PROJECTS

Jupeserver — Personal AI Orchestration System ***2025 – Present***

- End-to-end scheduled Python system (FastAPI + Google Cloud Functions) aggregating calendar, tasks, and notes (Google Calendar, MS Todo, org-mode files), routed through a custom multi-provider LLM abstraction with fallback logic (Anthropic, DeepSeek, OpenAI) to generate prioritized daily briefings and time-block plans. Google Sheets data layer with versioned schema migrations and OAuth via Google and Microsoft Graph APIs.

Probabilistic Geospatial Crop Mask Pipeline (Gro Intelligence) ***2019 – 2021***

- Productionized crop mask pipeline from analyst GEE prototype: 3,000+ batch export tasks per cycle, Celery harvest/transform with GCS checkpoint resilience, producing annual 30m floating-point per-crop probability rasters
- Key input to Gro's Crop Yield Forecast Model, outperformed USDA (Python, GEE, Celery, GDAL, AWS)

SKILLS

Languages: Python (expert), TypeScript/JavaScript, C++, SQL, C, Bash
AI/ML: LLM APIs (Anthropic, Gemini, OpenAI), PyTorch, TensorFlow, Claude Code, agentic workflows
Infrastructure: AWS, GCP, Docker, Kubernetes, Linux, PostgreSQL/PostGIS, Apache Solr, GraphQL, Firebase, Celery
Data: Pandas, NumPy, GDAL, Rasterio, Google Earth Engine, CUDA

EDUCATION

Swarthmore College — B.A. Computer Science *May 2017 | GPA: 3.7*

Minors: Mathematics (focus in Statistics and Probability) & Theater

Coursework: Distributed Systems, Probabilistic/Stochastic Modeling, Computer Vision, Linear Algebra, Real Analysis, Robotics, Multivariable Calculus, Statistics